

AGENEWTECH

M1642ZP

Smart Module Specification_V2.1



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About the Document

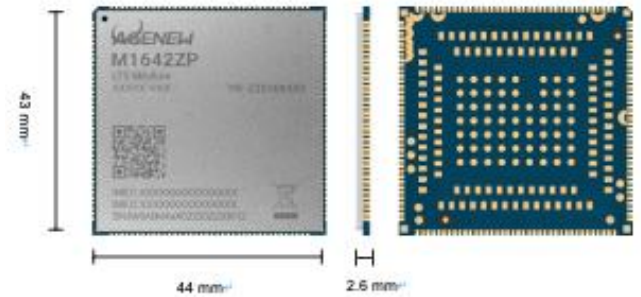
History

Version	Data	Author	Description
-	2023-12-26	Zhou Yuqiang	Initial
1.0	2023-12-28	Huang Zizhen	Controlled Copy
2.0	2024-05-05	Wu Danyi	Modified to a single-sided version definition
2.1	2026-03-23	Chen Yun	Change company name

M1642ZP

Multi-network Smart LTE module

Supports Wi-Fi & BT



M1642ZP is a multi-network Smart LTE Cat 7 module based on MediaTek's Octa-core MT8781 (ARM Cortex-A76+ARM Cortex-A55) processor launched by Wuxi Sunning Technology Co., Ltd, which is equipped with Android 13 operating system. Its powerful performance, neon engine processing, 2D/3D graphics acceleration processing capabilities, fully meet the needs of customers for high-speed and multimedia functions in industrial and consumer applications.

M1642ZP supports full coverage of various network standards, Wi-Fi 802.11a/b/g/n/ac and BT5.2 LE short-range communication. Not only that, M1642ZP also has audio and video I/O interfaces and rich GPIO interfaces, but also supports a variety of audio and video codecs, and the main screen supports up to 2K. At the same time, the positioning system of GPS/GLONASS/BDS/Galileo are supported by M1642ZP makes its positioning function fast and accurate.

M1642ZP is a SMT module with 321 pins, including 152 LCC pins and 169 LGA pins. With a size of only 43.0mm × 44.0mm × 2.6mm, it can be embedded in various M2M products through pads, making it widely used in smart cash registers, smart POS, tax control machines, security monitoring, vehicle-mounted equipment, high-end information collection equipment, intelligent robots, smart homes, smart hardware, industrial smart handheld equipment, unmanned aerial vehicles, high-end police or law enforcement equipment, smart intercom equipment, smart wearing and vending machines.

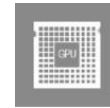


Main Advantages

- ✓ Comprehensive coverage of multi-network standard.
- ✓ Support Wi-Fi 802.11a/b/g/n/ac and BT5.2 short-range wireless communication
- ✓ Support GPS/GLONASS/BDS/Galileo satellite positioning
- ✓ Support multiple audio and video codecs
- ✓ The main screen supports up to 2000×1200
- ✓ Support 2K@30fps video recording and playback
- ✓ Possess audio and video input and output interfaces and rich GPIO interfaces



ARM A76+ A55
64-bit Processor
(Octa-core)



ARM G57 MC2®



Support Android 13



LTE Cat 7
Max 300Mbps (DL)
Max 75 Mbps (UL)



IEEE 802.11
a/b/g/n/ac



Bluetooth 5.2
(BR/EDR + BLE)



GPS/GLONASS/
BeiDou/Galileo



NEON
Multimedia
processing engine



LCC + LGA
encapsulation

LTE 4G	M1642ZP
Region/Operator	China/India/South Korea/Thailand/Japan/Australia/Taiwan/Brazil, Europe, Middle East, Africa, North America, Latin America
Operating system	Android 13
Memory	UFS 64G+LPDDR4X 32G
Module size (mm)	43.0 × 44.0 × 2.6
Packaging	LCC + LGA
Temperature range	
Operation temperature	Operating temperature: -20 °C ~ + 80 °C Storage temperature: -40 °C ~ + 90 °C
Frequency band information	
LTE-FDD	B1/B3/B5/B7/B8/B20/B28
LTE-TDD	B34/B38/B39/B40/B41(120M)
WCDMA	B1/B2/B5/B8
GSM/EDGE	B2/B3/B5/B8
WLAN	2.4/5GHz, 802.11a/b/g/n/ac, support AP and STA mode
Bluetooth	BT 5.2
GNSS	GPS/GLONASS/Beidou/Galileo
Authentication	
Operator Certification	Undetermined
Compulsory Certification	CE/SRRC*/NAL*/CCC*
Other Certifications	RoHS/REACH
Frequency Band Information	
LTE-FDD (Mbps)	300Mbps (DL)/75Mbps (UL)
LTE-TDD (Mbps)	240Mbps (DL)/75Mbps (UL)
DC-HSPA+ (Mbps)	42Mbps (DL)/11.5Mbps (UL)
WCDMA (Kbps)	384Kbps (DL)/384Kbps (UL)
EDGE (Kbps)	236Kbps (DL)/236Kbps (UL)
GPRS (Kbps)	85.6Kbps (DL)/85.6Kbps (UL)
Outstanding characteristics	
Five-mode communication	LTE: Cat7 2CA UMTS: Release 8
Bandwidth (MHz)	1.4-20MHz
Charging chip	●
OTG	●
SIM card detection	●
Dual sim dual standby	●
USB Firmware upgrade	●
OTA Firmware upgrade	●

Remarks: 1. *: In progress
2. ●: Supports

LTE Cat4 M1642ZP	
Electrical specification	
Operating Voltage	3.4~4.4 V, typical 3.8 V
Consumption (typical)	Undetermined
Interface	
LCD	Video codec: 2K@30fps; Wi-Fi video: encoding up to 2K@30fps; Decode up to 2K @30fps
Camera	3-set 4-lane MIPI CSI, supports three groups of cameras, and can support up to 32MP pixels
TP	Capacitive touch screen
Audio	3 analog inputs: 3 sets of analog microphone inputs
	3 analog outputs: speaker, earpiece, headphones
USB	USB 2.0, support OTG mode
(U)SIM	×2 (1.8V/2.95V)
UART	×2, for debugging and application
SD Card	×1
ADC	×2
I2C	×7, for control peripherals
SPI	×5
PWRKEY	Pull up internally to VBAT
GPIO	14 GPIO interfaces and 28 dedicated GPIOs (reusable)
Antenna	Main antenna, diversity antenna, GNSS antenna, Wi-Fi 2.4G&BT antenna, Wi-Fi 5G antenna